Billion-scale similarity search with GPUs

Department of Computer Science & Artificial Intelligence

Abstract

This paper investigates rag & retrieval with extensive empirical and theoretical analysis. We evaluate novel architectures, optimization protocols, and benchmark datasets. Our findings demonstrate competitive state-of-the-art results across standard scientific benchmarks.

1. Introduction

Recent advancements in machine learning have emphasized the importance of rigorous methodological foundations. In this work, we present systematic findings regarding Billion-scale similarity search with GPUs.

2. Technical Methodology & Empirical Findings

Continuing detailed technical derivations for Billion-scale similarity search with GPUs. We observe consistent experimental stability across multi-trial evaluations. The parameters exhibit robust convergence with bounded variance. Ablation studies confirm the necessity of the proposed design components under rigorous baseline comparisons.

3. Technical Methodology & Empirical Findings

Section topic: RAG & Retrieval
Question context: What is the difference between Inverted File (IVF) and Hierarchical Navigable Small World (HNSW) index structures in FAISS?
Key Finding: IVF clusters vectors into Voronoi cells for inverted list lookup, whereas HNSW creates a hierarchical graph for rapid graph traversal [Billion-scale similarity search with GPUs, p. 3].
Evidence passage: IVF partitions vector space into Voronoi cells using k-means clustering HNSW constructs a multi-layer proximity graph supporting logarithmic search time

4. Technical Methodology & Empirical Findings

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